

EXP NO – 19

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

AIM

To design and simulate an ultrasonic distance measurement system using an HC-SR04 sensor and Arduino UNO in Proteus. The system measures the distance of an object and displays the distance through the Serial Monitor.

PROGRAM

```
#define TRIG_PIN 9
#define ECHO_PIN 10

long duration;
float distance;

void setup()
{
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);

    Serial.begin(9600);
}

void loop()
{
    // Make sure trigger is LOW
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    // Send 10 microsecond trigger pulse
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    // Measure echo pulse
    duration = pulseIn(ECHO_PIN, HIGH);
```

```
Serial.print("Distance = ");
Serial.print(distance);
Serial.println(" cm");
```

$$\}$$

The screenshot shows the Proteus 8 Professional Schematic Capture window. The main workspace displays a circuit diagram with an Arduino Uno (ARD1) and a SONAR1 Ultrasonic Sensor. The Arduino Uno is connected to the sensor's VCC, GND, and Trig pins. The sensor's Echo pin is connected to a 1k resistor and a 5V source. The serial monitor is connected to the Arduino's TXD, RXD, and CTS pins. The left sidebar shows the 'DEVICES' list with 'ARDUINO UNO' and 'ULTRASONIC SENSOR' selected. The bottom status bar indicates 'ANIMATING: 00:00:02.356756 (CPU load 65%)'.

OUTPUT



